

Revision · The volume of prisms and cylinders

Cambridge insert: one formula for every prism — cross-section times length.

LESSON 65

1 × 40 MIN

GEOMETRY 8 – REVISION

STAGE 9 · 14.1

LESSON CLOCK

5'

14'

12'

5'

4'

What is a prism?	5 min
The prism formula	14 min
Cylinders	12 min
Units of volume	5 min
Homework	4 min

LEARNING OBJECTIVES

- Use $V = S_{base} \cdot h$ for any prism.
- Find the volume of a cylinder.
- Find a missing dimension from a given volume.
- Convert between units of volume and of capacity.

TEXTBOOK

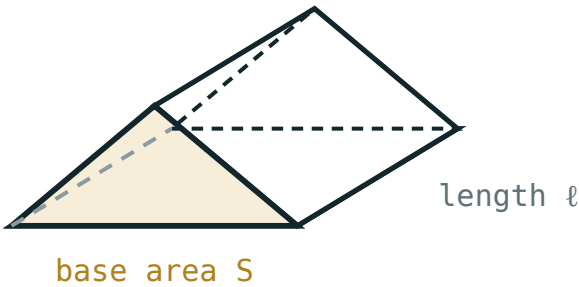
National	Annual revision
Cambridge	Learner's Book pp. 318–326
Workbook	Workbook 14.1

01

Explanation

One formula for all prisms

A **prism** is a solid with the same cross-section all the way along its length. Slice it anywhere across, and you get the same shape.



$$V = S \cdot \ell$$

A prism is a cross-section repeated along a length: the volume is the area of that cross-section times the length.

VOLUME OF A PRISM

$$V = S_{\text{cross-section}} \cdot h$$

A cylinder is a prism with a circular cross-section, so $V = \pi r^2 h$.

SOLID	CROSS-SECTION	V
cuboid	rectangle	abc
cube	square	a^3
triangular prism	triangle	$\frac{1}{2} a h_{\Delta} \cdot L$
cylinder	circle	$\pi r^2 h$

Learning one formula instead of four is the point of this lesson: find the area of the face that repeats, then multiply by how far it repeats.

Units and capacity

Volumes scale by the **cube** of the length factor:

$$1\text{ m}^3 = 1\,000\,000\text{ cm}^3 \mid 1\text{ cm}^3 = 1000\text{ mm}^3$$

Capacity uses litres, and the bridge between the two systems is worth memorising:

$$1\text{ litre} = 1000\text{ cm}^3 \mid 1\text{ m}^3 = 1000\text{ litres}$$

MIXED UNITS

A tank 80 cm by 1.2 m by 50 cm must be converted before multiplying. In metres: $0.8 \cdot 1.2 \cdot 0.5 = 0.48\text{ m}^3 = 480\text{ litres}$.

02 Worked examples

EXAMPLE 1 A triangular prism has a cross-section with base 6 cm and height 4 cm , and is 15 cm long. Find its volume.

$$1 \quad S = \frac{1}{2} \cdot 6 \cdot 4 = 12\text{ cm}^2$$

The cross-section.

$$2 \quad V = 12 \cdot 15$$

$$3 \quad = 180\text{ cm}^3$$

ANSWER

180 cm^3

EXAMPLE 2 A cylindrical tank has radius 50 cm and height 1.2 m . Find its capacity in litres.

$$1 \quad r = 0.5\text{ m}, h = 1.2\text{ m}$$

Convert first.

$$2 \quad V = \pi \cdot 0.25 \cdot 1.2$$

$$3 \quad = 0.3\pi \approx 0.942\text{ m}^3$$

$$4 \quad \approx 942\text{ litres}$$

$$1\text{ m}^3 = 1000\text{ l.}$$

ANSWER

$\approx 942\text{ litres}$

EXAMPLE 3 A cylinder has volume $500\pi \text{ cm}^3$ and height 20 cm . Find its radius.

① $500\pi = \pi r^2 \cdot 20$

② $r^2 = 25$
Divide by 20π .

③ $r = 5 \text{ cm}$

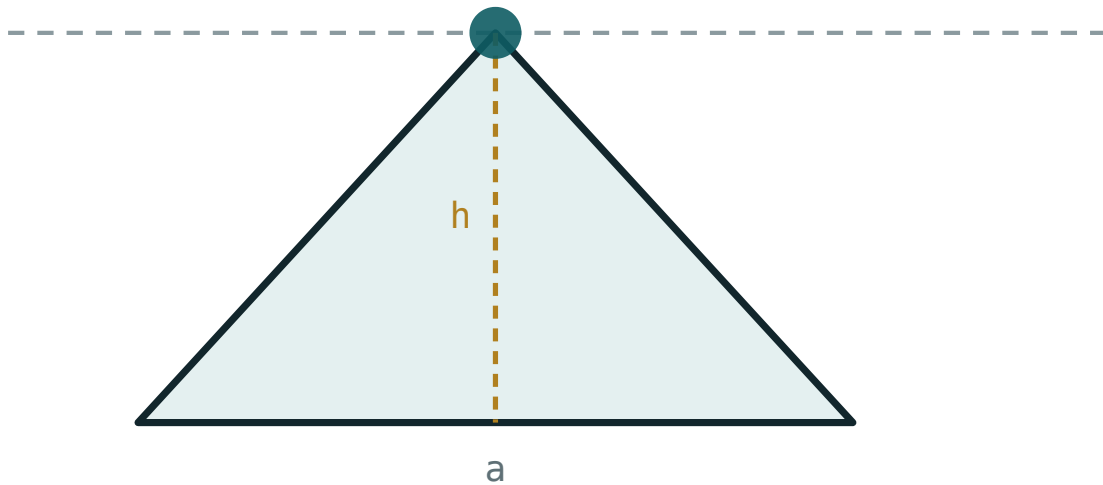
ANSWER

5 cm

03 Interactive model

Change the cross-section and see how the volume follows it directly.

The cross-section of a prism

base a **220**height h **120**area $\frac{1}{2}ah$ **13200**perimeter **545.6**

04 Terminology

Write these on the board at the start. Learners meet the national programme in Uzbek or Russian and the Cambridge papers in English — the three columns are the same idea.

ENGLISH	O'ZBEKCHA	РУССКИЙ
Prism	Prizma	Призма
Cross-section	Ko'ndalang kesim	Поперечное сечение
Cuboid	To'g'ri burchakli parallelepiped	Прямоугольный параллелепипед
Cylinder	Silindr	Цилиндр
Volume	Hajm	Объём
Capacity	Sig'im	Вместимость
Cubic metre	Kub metr	Кубический метр
Litre	Litr	Литр
Uniform cross-section	Bir xil kesim	Постоянное сечение

Stress the words in bold in the explanation above; ask for the Uzbek and Russian back before moving on.

05 Quick check**Check yourself**

A cuboid $3 \times 4 \times 5$ has volume:

A cylinder of radius 3 and height 10 has volume:

1 m^3 equals:

1 litre equals:

100 cm^3

1000 cm^3

10 cm^3

1 m^3

A prism has cross-section 15 cm^2 and length 8 cm . Its volume is:

23

120

60

7.5

06 Practice · 21 problems

Seven at each level. Set the easy row for everyone, the medium row for the main body of the class, and the hard row for those who finish early.

Easy · warm the idea up

7 PROBLEMS

1. Find the volume of a cuboid $2 \times 5 \times 6$.

ANSWER 60

2. Find the volume of a cube of side 4.

ANSWER 64

3. Find the volume of a cylinder with $r = 2$, $h = 5$ (in terms of π).

ANSWER 20π

4. A prism has cross-section 12 cm^2 and length 9 cm. Find its volume.

ANSWER 108 cm^3

5. Convert 3 m^3 to cm^3 .

ANSWER 3 000 000 cm^3

6. Convert 5000 cm^3 to litres.

ANSWER 5 l

7. Find the volume of a triangular prism: cross-section 10 cm^2 , length 7 cm.

ANSWER 70 cm^3

Medium · the standard question

7 PROBLEMS

1. Find the volume of a triangular prism with base 8, triangle height 5, length 12.

ANSWER 240

2. Find the volume of a cylinder with $r = 7$, $h = 10$ ($\pi \approx 3.14$).

ANSWER ≈ 1538.6

3. A cuboid has volume 240 and base 8×5 . Find its height.

ANSWER 6

4. A cylinder has volume 100π and radius 5. Find its height.

ANSWER 4

5. A tank is 60 cm by 40 cm by 50 cm. Find its capacity in litres.

ANSWER 120 l

6. Convert 0.75 m^3 to litres.

ANSWER 750 l

7. A cube has volume 125 cm^3 . Find its side and its surface area.

ANSWER 5 cm, 150 cm^2

Hard · stretch and reason

7 PROBLEMS

1. A cylinder has volume $1000\pi \text{ cm}^3$ and height 40 cm. Find its radius and diameter.

ANSWER $r = 5 \text{ cm}$, $d = 10 \text{ cm}$

2. A swimming pool is 25 m by 10 m, 1.2 m deep at one end and 2 m at the other, sloping evenly. Find its volume.

ANSWER trapezium cross-section $\frac{1.2 + 2}{2} \cdot 25 = 40 \text{ m}^2$, so $V = 400 \text{ m}^3$

3. A pipe has outer radius 5 cm, inner radius 4 cm and length 2 m. Find the volume of metal in it.

ANSWER $\pi(25 - 16) \cdot 200 = 1800\pi \approx 5655 \text{ cm}^3$

4. If all lengths of a prism are doubled, what happens to its volume?

ANSWER It becomes 8 times as large.

5. A cylindrical tank of radius 40 cm is filled at 20 litres per minute. How long to fill 1 m of its height?

ANSWER $V = \pi \cdot 0.16 \cdot 1 \approx 0.503 \text{ m}^3 = 503 \text{ l}$, so $\approx 25 \text{ minutes}$

6. Two similar cylinders have heights 6 and 9. The smaller has volume 48π . Find the larger.

ANSWER $k = 1.5$, $k^3 = 3.375$, so 162π

7. Explain why $V = S \cdot h$ works for every prism but not for a cone or a pyramid.

ANSWER A prism has the same cross-section at every height, so the volume is that one area repeated; a cone or pyramid tapers, its cross-section shrinking towards the apex, and its volume is only a third of the matching prism.

07 Homework

Homework — 5 problems

Cambridge Stage 9, Unit 14.1. Convert to one unit before multiplying.

- Find the volume of a cuboid $4 \times 7 \times 9$.
- Find the volume of a cylinder with $r = 6$ and $h = 12$ (in terms of π).

3. *A triangular prism has base 10, triangle height 6, length 15. Find its volume.*
4. *A tank is 80 cm by 50 cm by 60 cm. Find its capacity in litres.*
5. *A cylinder has volume 288π and radius 6. Find its height.*